G17 RESEARCH PROJECT

Preprocessing notebook

A detailed walkthrough of how the ultrasound datasets are prepared

The notebook turns three datasets into one reliable training source

Each section solves one problem before the model sees any image.

Setup

Install libraries, define paths and keep adjustable values in one configuration object.

Prepare

Load BUSI, BUS-BRA and BUS-UCLM; match every image with its mask.

Export

Audit, split, enhance and upload compact Parquet shards to Hugging Face.

Configuration keeps the whole run consistent

The CFG object stores values that may change between experiments.

```
class Config:
    SEED = 42
    SRAD_ITERATIONS = 30
    SRAD_STEP = 0.20
    SRAD_DECAY = 0.125
    CLAHE_CLIP_LIMIT = 2.0
    CLAHE_GRID = (8, 8)
    OUTER_FOLDS = 5
    INNER_FOLDS = 5
```

What it means

Instead of searching through many cells, the team changes settings once. The seed makes the run repeatable. SRAD and CLAHE control enhancement; the fold settings control grouped splitting.

All loaders return the same table format

The original datasets use different folders and naming rules, so the notebook standardises their metadata.

Common columns

dataset, image_id, patient_id, class_label, image_path and mask_path.

Why this matters

The later audit, split and export functions can process every dataset with the same code.

Outcome

A single development_index table represents all resolved image–mask pairs.

Each ultrasound must be paired with its correct mask

The loader resolves image and mask filenames before preprocessing.

```
records.append({
  "dataset": "BUSI",
  "image_id": image_id,
  "patient_id": patient_id,
  "class_label": label,
  "image_path": str(image_path),
  "mask_path": str(mask_path),
})
```

What it means

The image is the model input. The mask is the correct pixel-level answer. An incorrect pair would teach the network the wrong lesion location, so unresolved pairs are rejected.

The audit removes broken pairs before splitting

Geometry is checked while the complete dataset is still together.

```
image = read_gray(row.image_path)
mask  = read_mask(row.mask_path)

if image.shape != mask.shape:
    rejected.append({
      "image_id": row.image_id,
      "reason": "shape_mismatch"
    })
```

What it means

This catches the earlier HESN_003-type error before export. A mask cannot label the right pixels if its height and width do not match the image. Rejected items are written to an audit report.

Grouped splitting reduces information leakage

Known patient identities and duplicate-image groups are kept within one split.

Train

Used repeatedly to update the model weights.

Validation

Used during training to monitor progress and select the best state.

Internal test

Kept unseen until final evaluation.

Important caveat

BUSI lacks patient IDs, so its grouping is image-level; UCLM and BUS-BRA use available or derived patient groups.

Robust scaling prepares unusual brightness ranges

Extreme dark and bright values are clipped before enhancement.

```
lo, hi = np.percentile(image, (0.5, 99.5))
image = np.clip(image, lo, hi)
image = (image - lo) / (hi - lo)
return image.astype(np.float32)
```

What it means

Different machines can produce different brightness ranges. Percentile scaling reduces the influence of rare extreme pixels and converts the scan to a stable 0–1 range.

SRAD reduces speckle without simply blurring edges

Speckle-reducing anisotropic diffusion updates the image gradually.

```
for _ in range(iterations):
    north, south, east, west = gradients(image)
    c = diffusion_coefficient(image, gradients)
    image += 0.25 * (
        cN*north + cS*south +
        cE*east  + cW*west
    )
```

What it means

Ultrasound speckle looks like grain. Ordinary smoothing can erase lesion boundaries. SRAD diffuses strongly in uniform regions and less strongly near structural edges.

CLAHE improves contrast in small regions

Contrast enhancement is applied after SRAD.

```
clahe = cv2.createCLAHE(
    clipLimit=CFG.CLAHE_CLIP,
    tileGridSize=CFG.CLAHE_GRID
)
enhanced = clahe.apply(despeckled_u8)
```

What it means

CLAHE divides the scan into tiles and improves each tile locally. The clip limit prevents extreme amplification. This makes faint boundaries clearer without treating the entire image as one brightness region.

One function produces every saved image version

The same transformation is reused for every accepted image.

```
def preprocess_image(raw):
    scaled = robust_unit_scale(raw)
    srad_image = srad(scaled)
    srad_u8 = to_uint8(srad_image)
    enhanced = apply_clahe(srad_u8)
    return srad_u8, enhanced
```

What it means

The sequence is fixed: scale → SRAD → CLAHE. Reusing one function prevents slightly different preprocessing rules across datasets or splits.

Previewing is a scientific safety check

The preview cell samples each source dataset and displays the processing stages side by side.

Raw

Confirms the correct ultrasound was loaded.

SRAD

Checks that speckle decreased without destroying boundaries.

Enhanced

Checks local contrast after CLAHE.

Mask

Confirms the annotation still aligns with the scan.

Export creates compact Hugging Face dataset shards

Each record stores the enhanced image, its mask and identifying metadata.

```
DatasetDict({
  "train": train_dataset,
  "validation": val_dataset,
  "test": test_dataset
}).push_to_hub(
  "aee4/G17-preprocessed-dataset"
)
```

What it means

Parquet/Arrow shards bundle many examples into a few large files. Kaggle therefore downloads a few shards instead of making thousands of separate image requests and hitting the Hub resolver limit.

Four exported folders preserve each

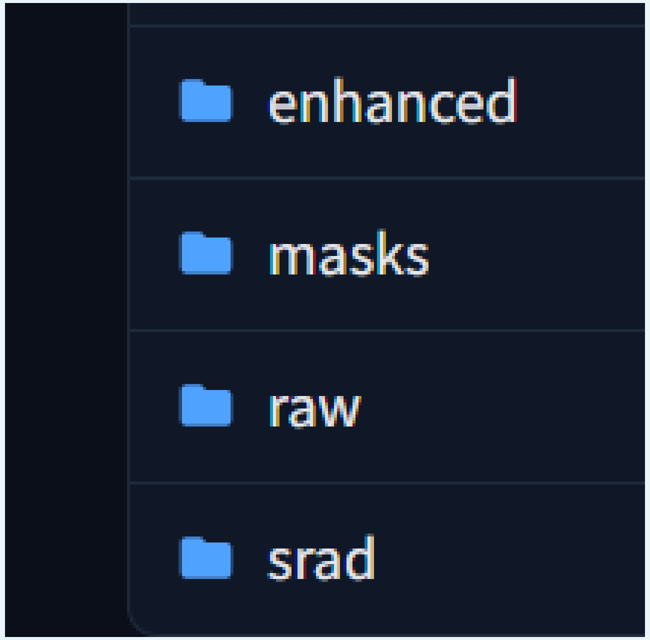
Folder meanings
raw — original scan

srad — despeckled scan

enhanced — SRAD + CLAHE result

masks — lesion annotations

Training uses enhanced images with their matching masks.



The preprocessing notebook ends with training-ready evidence

The result is more than a folder of cleaned pictures.

Reproducible data

Configuration and manifest record exactly what was processed.

Protected evaluation

Patient-aware splits keep the internal test independent.

Efficient access

Parquet shards load quickly in the training notebook.

Direct hand-off

Enhanced image + mask pairs are ready for segmentation training.